



# A Long-Term Regional Assessment of the CONUS404 Bulk Parameterization in Urban Areas



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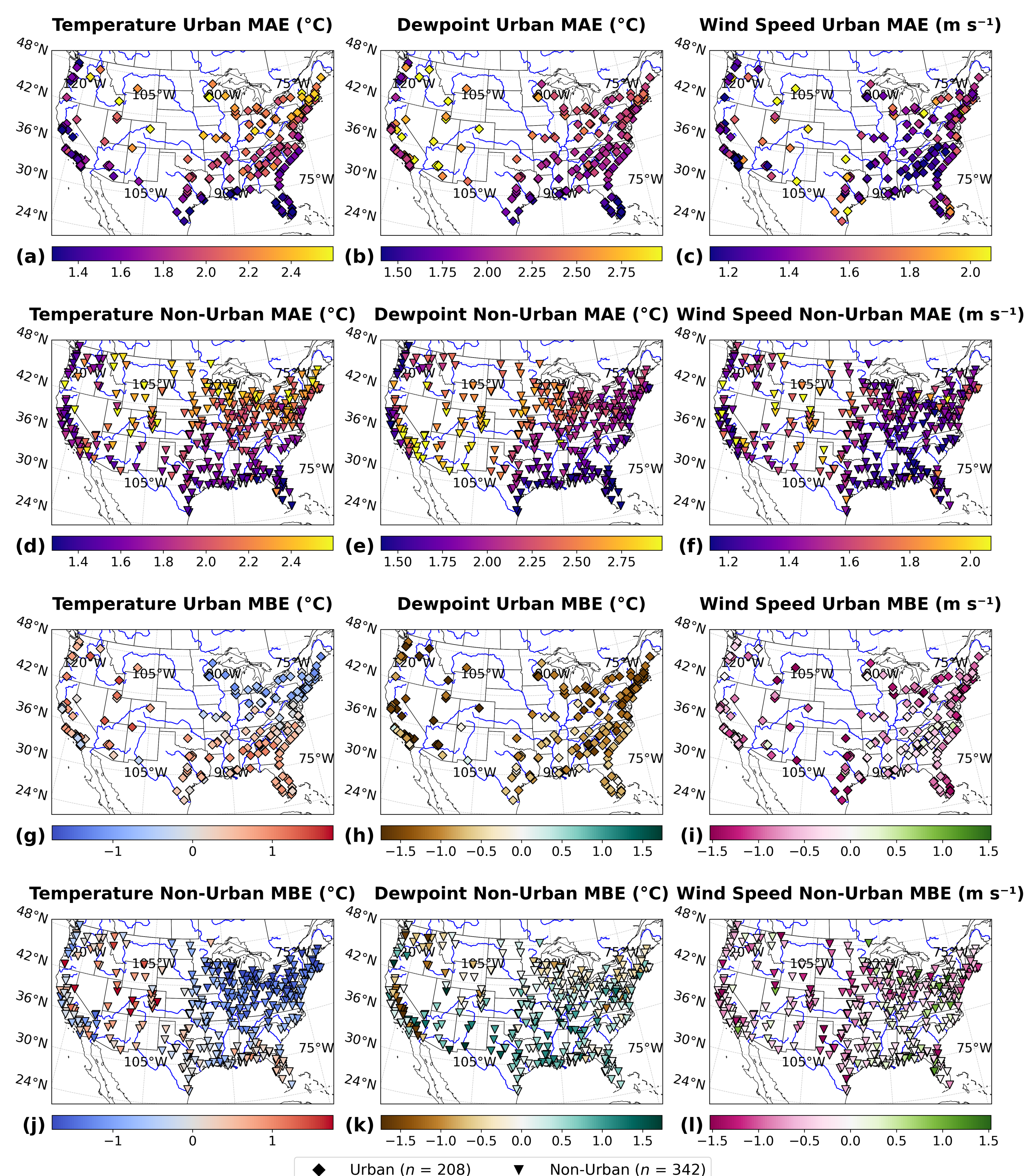
## i) Background

- CONUS404 is one of the highest resolution (4-km) convection-permitting reanalysis datasets, which could allow the model to be a critical tool in understanding unique urban dynamics (Rasmussen et al., 2023).
- The accuracy of the CONUS404 bulk approach in simulating atmospheric conditions in urban areas has been underexplored.
- Accurate representations of urban features in numerical weather models are crucial for accurately simulating complex interactions between cities and commonly observed atmospheric processes (e.g. urban heat island).
- To date, there have been no comprehensive continental-scale, spatial assessments of CONUS404's errors and biases in urban grids using hourly, station level observations.
- Our assessment provides a unique direct comparison of performance in urban grids to non-urban grids on a continental scale—a feat past studies have lacked. As a result, these results have implications for improving the accuracy of simulated urban hydrometeorological and climate conditions at continental and regional scales.

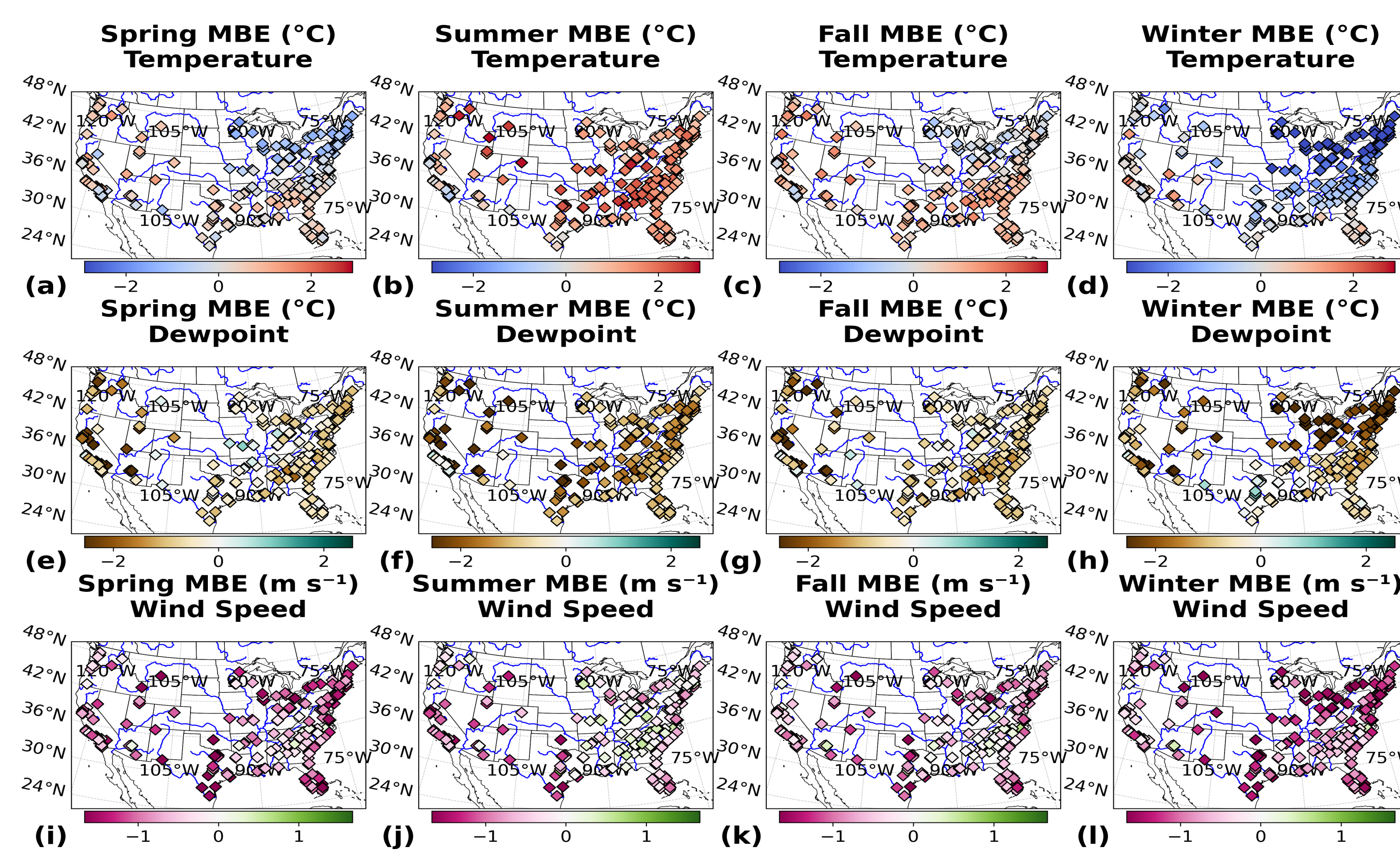
## ii) Methodology

- Utilized the gap-filled Comprehensive Hourly Urban Weather Database (CHUWD-H v1.0) to derive hourly station-level, observed data.
- Retrieved 10-years of CONUS404 2-m temperature, 2-m dewpoint, 10-m zonal wind component, and 10-m meridional wind component data at 550 sites from the National Center for Atmospheric Research (NCAR) Research Data Archive.
- Calculated mean absolute error (MAE)(tests accuracy) and mean bias error (MBE)(systematic bias) as CONUS404 – CHUWDH
- Determined urban and non-urban grids using the MODIS derived dominant land use category from 2010
- Applied and mapped seasonal analyses to gauge temporal performance
- Quantified the regionality of the model's errors and biases using the nine NOAA Climate Regions (Karl and Koss, 1984)

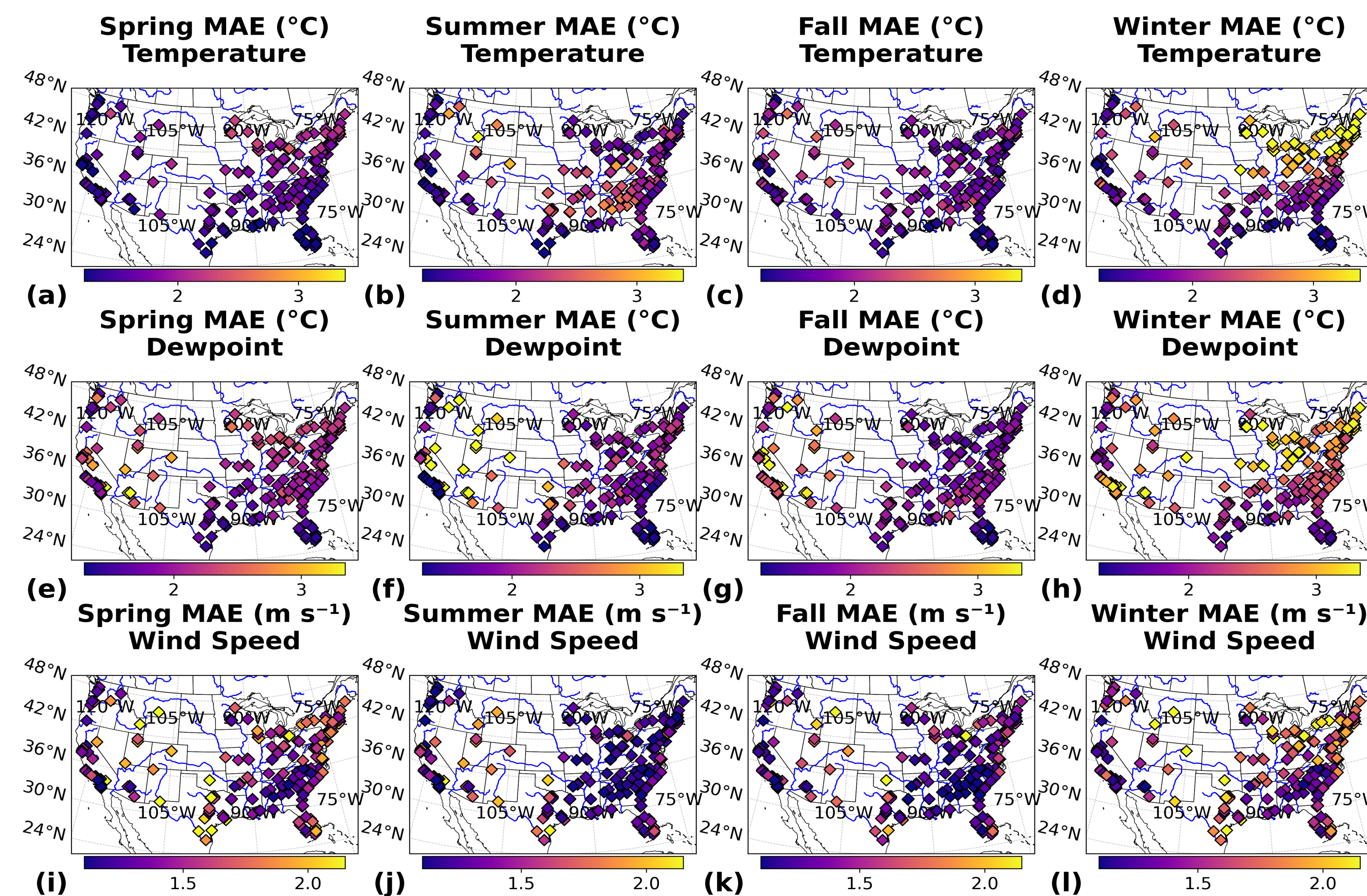
## iii) Regional Analysis



**Fig. 2:** Spatial patterns of (a-f) MAE and (g-l) MBE based on the comparison between hourly (a, d, g, and j) temperature, (b, e, h, and k) dewpoint, and (c, f, i, and l) simulated by CONUS404



**Fig. 3:** Seasonal spatial patterns of urban MBE based on the comparison between hourly (a-d) temperature, (e-h) dewpoint, and (i-l) wind speed simulated by CONUS404



**Fig. 4:** Seasonal spatial patterns of urban MAE based on the comparison between hourly (a-d) temperature, (e-h) dewpoint, and (i-l) wind speed simulated by CONUS404

## iv) Key Points

- Temperature, dewpoint, and wind speed in urban grids displayed lower accuracy in the South, Ohio Valley, Northeast, and Southeast.
- Temperature in urban grids showed statistically significant poorer performance during the summer and a statistically significant warm MBE.
- Winter (Northeast) was the worst performing season, for all simulated variables.
- Temperature biases in urban and non-urban grids are regionally constrained.
- Strong spatial and temporal dewpoint dry biases in urban grids, especially in the winter.
- Magnitude of wind speed underestimation is stronger in urban grids and is persistent at all temporal scales.

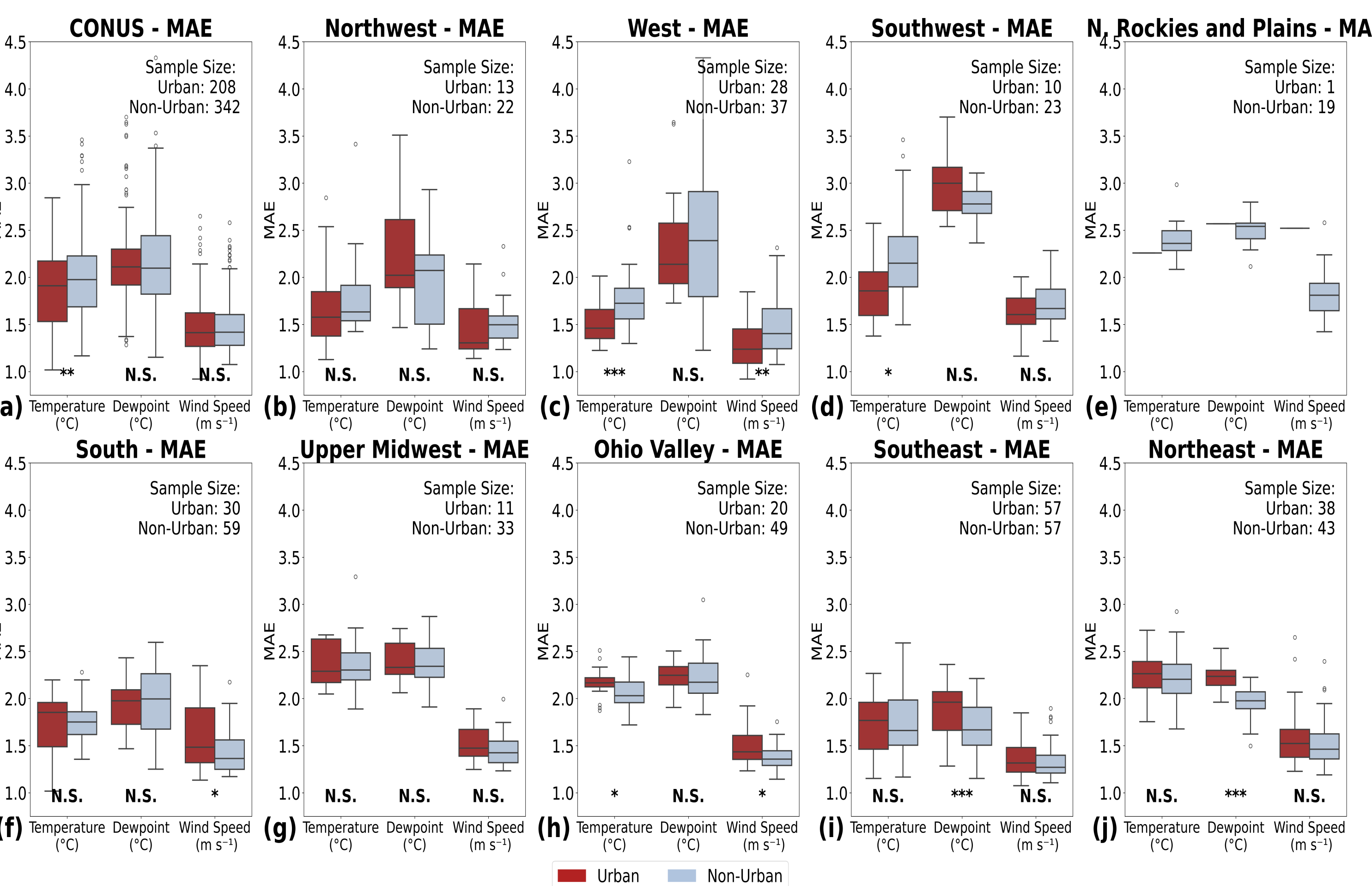
## v) References

Karl, T. R., and Koss, W. J. (1984). Regional and national monthly, seasonal, and annual temperature weighted by area, 1895–1983. National Climatic Data Center Historical Climatology Series, 4(3), 38 pp.

Rasmussen, R., Chen, F., Liu, C.H., Ikeda, K., Prein, A., Kim, J., Schneider, T., Dai, A., Gochis, D., Dugger, A., Zhang, Y., Jaye, A., Dudhia, J., He, C., Harrold, M., Xue, L., Chen, S., Newman, A., Dougherty, E., Abolafia-Rosenzweig, R., Lybarger, N.D., Viger, R., Lesmes, D., Skalak, K., Brakebill, J., Cline, D., Dunne, K., Rasmussen, K., Miguez-Macho, G., 2023b. CONUS404: The NCAR-USGS 4-km long-term regional hydroclimate reanalysis over the CONUS. Bull. Am. Meteorol. Soc. 104, E1382–E1408. <https://doi.org/10.1175/BAMS-D-21-0326.1>

## vi) Acknowledgements

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**Fig. 1:** MAE results between urban (red) and non-urban (gray) grids. Welch's t-test was performed in eight regions, to determine statistical significance of differences in mean MAE.